DSA REFERENCE

Bit Manipulation Failure Modes

Common wrong turns, debugging cues, and complexity traps.

Use the binary representation directly for compact state, parity, subsets, and cancellation properties.

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DSA / BIT-MANIPULATION

DSA

Bit Manipulation Failure Modes

CORE IDEA

Use the binary representation directly for compact state, parity, subsets, and cancellation properties.

Recognition cues

+ power of two
+ single unique value
+ subset masks
+ flags or parity

Implementation sequence

+ state signed-width assumptions
+ use masks with names
+ apply x & (x-1) to clear the lowest set bit
+ test zero separately

Invariant

Each bit has one documented meaning, and operations preserve all unrelated bit positions.

Worked reasoning

XOR cancels paired values because x xor x is zero and zero xor y is y.

Complexity

Often O(n) over values or O(number of machine bits), with O(1) extra space.

Failure modes

- [ ] operator precedence
- [ ] negative-number representation
- [ ] shifting beyond width
- [ ] using cleverness without explanation

Practice ladder

Bit Manipulation Failure Modes: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Single Number**	State the cue, invariant, edge case, and proof before implementation.
2	**Number of 1 Bits**	State the cue, invariant, edge case, and proof before implementation.
3	**Counting Bits**	State the cue, invariant, edge case, and proof before implementation.
4	**Reverse Bits**	State the cue, invariant, edge case, and proof before implementation.
5	**Sum of Two Integers**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

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